MIL-P-10480D (MU)
5 MARCH 1973
SUPERSEDING
MIL-P-10480C (MU)
30 JUNE 1965
W/AMENDMENT 1
30 NOVEMBER 1966

MILITARY SPECIFICATION

PLUNGER, DELAY, M1
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of plunger designated as Plunger, Delay, M1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - *Fuze and Fuze Components*, Environmental and Performance Tests for
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

ARMY

8797864 - Delay Plunger, M1, Assembly
8858565 - Box, Packing, Ammunition for Delay Plunger, M1
8858566 - Carton, Packing, Ammunition for Delay Plunger, M1

PUBLICATIONS

ARMY

ET-8797864 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Delay Plunger. - The delay plungers shall comply with all requirements specified on Drawing (dwg.) 8797864 and with all requirements specified in applicable specifications.

3.3 Jolt and Jumble. - The delay plunger assemblies shall comply with the requirements of MIL-STD-331, Tests 101.1 and 102.1.

3.4 Drop. - The delay plunger assemblies shall have no explosive elements explode when subjected to a drop of not less than three (3) feet or more than three and one half (3 1/2) feet.

3.5 Arming. - The delay plunger assembly shall have no parts fail and shall move to the armed position when rotated at a speed of not less than 2075 revolutions per minute (rpm) nor more than 2125 rpm. The armed position is that position whereby both centrifugal plunger pins are fully retracted into the body cavity and locked in that position by the centrifugal plunger pin lock.

3.6 Non-Arming. - The delay plunger assembly shall have no parts fail and shall remain in the unarmed position when rotated at a speed of not less than 1300 rpm nor more than 1325 rpm. The unarmed position is that position whereby the distance between the centrifugal plunger pins is less than 0.115 inch.

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3.7 Functioning. - The delay plunger assembly shall function satisfactorily as evidenced by the subsequent detonation of the M17 Detonator when tested as specified in 4.3.5.

3.8 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provisions. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated

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by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification or drawing not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Unless otherwise approved by the Contracting Officer, the inspection lot size of end items deliverable under the contract shall not be less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government Inspector. Each lot shall contain:

a. Parts and parts assemblies of one lot interfix number from one manufacturer.
b. Delay elements of one lot interfix number from one manufacturer with one delay time.

4.2.2 Examination. - The examinations listed in the following sub-paragraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235

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may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Assembly, Less Firing Pin Assembly (see dwg. 8797864).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Delay element assembly not staked	Visual	01001
2.	Spring retainer missing	Visual	01002
Major:	AQL 0.65 Percent		
101.	Delay element assembly missing or incorrect	Visual	01003
102.	Top of delay element not flush to plunger body face	Gage	01004
103.	Evidence of poor workmanship (see 3.8)	Visual	01005
Minor:	None defined.		

4.2.2.2 Assembly, After Completion of Non-Arming Test (see dwg. 8797864).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Centrifugal plunger pin in armed position	Visual	02001
2.	Plunger restraining spring missing.	Visual	02002
3.	Plunger support missing	Visual	02003
Major:	AQL 1.0 Percent		
101.	Height of guide pin above plunger body	Gage	02004
102.	Diameter, maximum (max.)	Gage	02005

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103.	Guide pin loose	Manual	02006
104.	Lock check washer missing or improperly positioned	Visual	02007
105.	Centrifugal plunger pin lock missing	Visual	02008
Minor:	AQL 1.5 Percent		
201.	Marking misleading or unidentifiable	Visual	02009
202.	Firing pin assembly dented	Visual	02010
203.	Evidence of poor workmanship (see 3.8)	Visual	02011

4.2.2.3 Pin, Guide (see dwg. 9204747).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Length	Gage	03001
102.	Diameter	Gage	03002
Minor:	AQL 1.0 Percent		
201.	Chamfer missing	Visual	03003
202.	Evidence of poor workmanship (see 3.8)	Visual	03004

4.2.2.4 Carton, Sealed Packing (see dwg. 8858566).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Assembly damaged so that contents are exposed or liable to become exposed	Visual	04001
102.	Sealing strip incomplete, badly wrinkled or not in contact for at least 1/2 inch around entire carton	Visual	04002

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Minor:	AQL 1.0 Percent		
201.	Contents move when carton is shaken	Manual	04003
202.	Marking misleading or unidentifiable	Visual	04004

4.2.2.5 Bag, Sealed Barrier (see dwg. 8358566, when applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Waterproofness destroyed by separation of heat seal or rip, tear or puncture	Visual	05001
Minor:	None defined.		

4.2.2.6 Box, Packing, Wood, Sealed (see dwg. 8858565).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 Percent		
101.	Hardware missing, broken or loose	Visual-Manual	06001
102.	Strapping missing, broken or loose	Visual-Manual	06002
103.	Contents exposed	Visual	06003
104.	Board broken or split	Visual	06004
105.	DOD symbol misleading or unidentifiable	Visual	06005
Minor:	AQL 2.5 Percent		
201.	Hardware improperly engaged	Visual-Manual	06006
202.	Handle missing or insecure	Visual-Manual	06007
203.	Contents loose (when box is tilted)	Manual	06008
204.	Strapping improperly assembled	Visual	06009

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205.	Marking misleading or unidentifiable	Visual	06010
206.	Car seal missing, unsealed or improperly positioned	Visual	06011

4.2.3 Testing

4.2.3.1 Arming (see 3.5) - Major Defect - Code No. 02014. - This test shall be performed 100 percent. Any delay plunger which fails to comply with the requirements of 3.5 shall be classed defective and removed from the lot. The tests shall be performed as specified in 4.3.3 using equipment in accordance with 4.2.4.

4.2.3.2 Non-Arming (see 3.6) - Critical Defect - Code No. 02015. - This test shall be performed 100 percent. Any delay plunger which fails to comply with the requirements of 3.6 shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.4 using equipment in accordance with 4.2.4.

4.2.3.3 Jolt and Jumble (see 3.3) - Critical Defect - Code No. 02016.- Beginning with the first lot produced and continuing until three consecutive lots have complied with the specified requirements, 36 loaded delay plunger assemblies shall be selected from each lot. Thereafter, 12 delay plunger assemblies shall be selected from each lot. The same assemblies subjected to the jolt test shall be subjected to the jumble test. The test shall be performed as specified in 4.3.1 using equipment in accordance with 4.2.4. If one or more delay plunger assemblies fail to comply with the specified requirements, or if one or more plungers arm, function or cause explosive breakup (dusting), the lot shall be rejected.

4.2.3.4 Drop (see 3.4) - Critical Defect - Code No. 02017. - This test shall be performed 100 percent. The test shall be performed as specified in 4.3.2 using equipment in accordance with 4.2.4. Delay plunger assemblies which fail this test shall be removed from the lot.

4.2.3.5 Functioning (see 3.7) - Major Defect - Code No. 02018. - Beginning with the first lot produced and continuing until three consecutive lots have been accepted, the sampling plan shall be in accordance with Standard MIL-STD-105 using Code Letter L and an AQL of 1.0 percent. Thereafter, the sampling plan for this test shall be in accordance with MIL-STD-105 using Code Letter J at an AQL of 1.0 percent.

4.2.3.6 Check Test for Deterioration of Delay Element Assembly (see MIL-D-46486A or MIL-D-48116) - Major Defect - Code No. 07001. - If the total time between original acceptance of any delay element assembly lot and the assembly of that lot into the delay plunger assemblies exceeds two years,

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or if the delay element assemblies have been subjected to adverse conditions, however brief, at any time since previous tests, the delay element assembly lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-D-46484A or MIL-D-48116 immediately before the delay element assembly lot is assembled into the delay plunger assemblies. This test shall be performed by the contractor on delay element assemblies selected by the Government Inspector at the facility assembling the delay element assemblies into the delay plunger assemblies (see 6.4).

4.2.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed in this section is identified by Equipment Tabulation ET-3797864 and under the "Method of Inspection" headings in 4.2.2 and 4.2.3 (see 6.3).

4.2.4.1 Government Rights to Documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Government Use of Contractor's Inspection and Test Equipment.- The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test Methods and Procedures

4.3.1 Jolt and Jumble. - The delay plunger assemblies shall be subjected to the jolt test specified in Standard MIL-STD-331, Test 101.1. After completion of the jolt test and prior to disassembly for inspection purposes, the same delay plunger assemblies shall be subjected to the jumble test specified in Standard MIL-STD-331, Test 102.1. Examination shall then be made to determine compliance with 3.3.

4.3.2 Drop. - The unarmed delay plunger assembly shall be placed in a specified fixture, with the plunger body facing up, and allowed to drop down a tube not less than 3.0 feet nor more than 3.5 feet. Observation shall be made to determine compliance with 3.4.

4.3.3 Arming. - The delay plunger assemblies shall be subjected to the arming test in accordance with all of the following requirements:

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a. The delay plunger assembly, in the unarmed position, shall be rotated in a vertical position with the open end of the housing in the upward position.

b. The acceleration imparted to the delay plunger assembly shall be as nearly uniform as practicable for each assembly under test.

c. No assembly shall be rotated at a speed exceeding 2125 rpm.

d. Total time from starting to stopping of the plunger assembly shall not exceed five (5.0) seconds.

e. Examination shall be made to determine compliance with 3.5.

f. Upon completion of the test, the assembly shall be restored manually to the unarmed position.

4.3.4 Non-Arming. - This test shall follow the arming test and shall be performed in accordance with 4.3.3, except that no assembly shall be rotated at a speed less than 1300 rpm, nor more than 1325 rpm. When using equipment which requires a visual inspection for the non-arming condition, the total time from starting to stopping of the rotation of the plunger body assembly shall not be less than three (3) seconds. Examination shall be made to determine compliance with 3.6.

4.3.5 Functioning. - The delay plunger assembly, in the armed position, shall be placed in a holding fixture which simulates the delay portion of the M557 fuze. The M17 Detonator shall be located below the straight through flash hole of the M1 plunger body at a distance of 0.150 inches. A weight not exceeding eight (8) ounces shall fall from a vertical height not exceeding twelve (12) inches to impact the housing of the assembly. Observation shall be made to determine compliance with 3.7. The test fixture shall be as specified in 4.2.4. Assemblies subjected to this test shall not be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C. - Packaging for Plunger, Delay, M1 shall be in accordance with dwg. 8858566.

5.2 Packing

5.2.1 Level C. - Packing for Plunger, Delay, M1 shall be in accordance with dwg. 8858565.

5.3 Marking. - Marking shall be in accordance with the applicable packing drawing.

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6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data Cards. - Data cards shall be perpared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Code Numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers on the Equipment Lists referenced on the Equipment Tabulation. Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with

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SIE designs if not otherwise covered thereby (see 6.3.1.c). Designs shall be of the category and form (per MIL-D-1000 specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-F, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Cost of Check Test. - The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of delay element assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the delay element assemblies or to the contractor assembling the delay element assemblies into the delay plungers, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the delay element assembly lot was under the jurisdiction of either contractor.

6.5 Check Test for Deterioration of Plunger Assembly. - The following test is designed to prevent the loading into a major ammunition item of delay plunger assemblies which have deteriorated. The test shall be performed on each lot of delay plunger assemblies which have been stored under normal conditions for more than two (2) years, or under adverse conditions for any period of time. If the specification for the ammunition item which this delay plunger assembly is loaded does not require that the check test for deterioration be performed, the Contracting Officer shall insert a provision for this check test in the contract.

6.5.1 Functioning (see 3.7). - The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using Code Letter M and an AQL of 2.5 percent. The test shall be performed as specified in 4.3.5 using

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equipment in accordance with 4.2.4. Any delay plunger assembly which fails to comply with the specified requirement shall be classed defective. Rejection shall be in accordance with Standard MIL-STD-105.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1375-A-016

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